Resistance of a Wire

Exam code: 0625 & 0972

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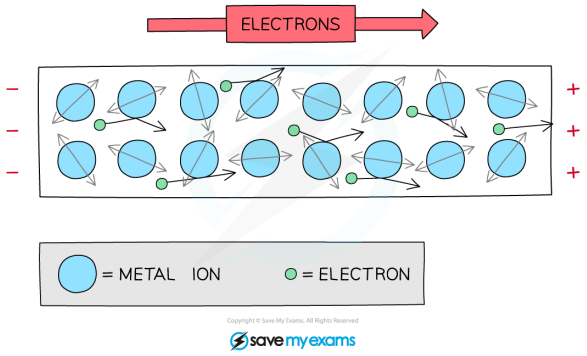
Resistance of a wire

- As electrons pass through a wire, they **collide** with the metal ions in the wire
- These collisions transfer energy away from the kinetic store of the electrons, which causes them to **slow down**
- The energy from the electrons is transferred to the kinetic store of the vibrating metal ions
  - This causes the vibration of the ions to increase (increased temperature)
  - As the vibration of the ions increases, the more the electrons collide with them (increased resistance)

Electron collisions in a metal wire



Electrons collide with metal ions, which resist their flow

- If the wire is longer, each electron will collide with more ions, and so there will be more resistance:
  - **The longer a wire, the *greater* its resistance**
- If the wire is thicker (greater diameter) there is more space for the electrons and so more electrons can flow:
  - **The thicker a wire, the *smaller* its resistance**

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Did this video help you? Yes No

Proportionality relationships for electrical conductors

Extended tier only

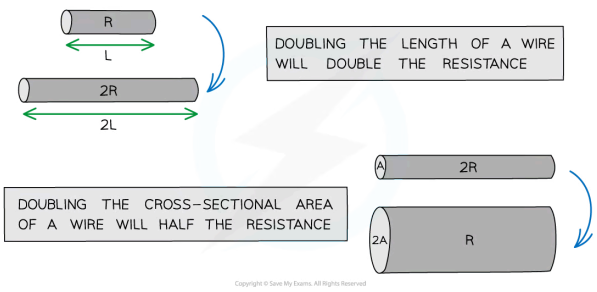
- The relationship between resistance, length and cross-sectional area can be represented mathematically
- Resistance is **directly** proportional to **length**
  - Doubling the length will double the resistance and vice versa

$$R \propto L$$

- Resistance is **inversely** proportional to **cross-sectional area** (width, or thickness)
  - Doubling the cross-sectional area will halve the resistance

$$R \propto \frac{1}{A}$$

Effect of length and cross-sectional area on resistance



The mathematical relationship between length and width of the wire and the resistance

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